$\boxed{\text{1st Iteration}}$ $W_1 = W_1 + \alpha\,(\text{label} - y\,\text{Pred})\,X_1$ no:

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Sample 1:

$Z = (-0.5)(0.2) + (0.1)(0.5) + (0.4)(0.7) + 0.2 = 0.43$

$\hat{y} = \text{sign}(0.43) = 1$ (correct)

Sample 2:

$Z = 0.42$ $\hat{y} = \text{sign}(0.42) = 1$ (missclassified)

$W_1 = -0.5 + 0.01(-2)\,0.1 = -0.502$

$W_2 = 0.094$ $W_3 = 0.388$ $b = 0.18$

Sample 3:

$Z = 0.132$ $\hat{y} = \text{sign}(0.132) = 1$ (correct)

Sample 4:

$Z = -0.502(0.6) + (0.094)(0.4) + (0.9)(0.388) + 0.18$
$= 0.2656$

$\hat{y} = \text{sign}(0.2656) = 1$

(missclassified)

$W_1 = -0.502 + 0.01(-2)(0.6) = -0.514$

$W_2 = 0.094 + 0.01(-2)(0.4) = 0.086$

$W_3 = 0.388 + 0.01(-2)(0.9) = 0.37$

$b = 0.18 + 0.01(-2) = 0.16$

Date: ______ No. ______

2nd Iteration

Sample 1:

$Z = 0.3592$ $\hat{y} = sign(0.3592) = 1$

(correct)

Sample 2:

$Z = 0.3564$ $\hat{y} = sign(0.3564) = \cancel{1}\ 1$

(missclassified)

$W_1 = -0.514 + 0.01(-2) \times 0.1 = -0.516$
$W_2 = 0.086 + 0.01(-2) \times 0.3 = 0.08$
$W_3 = 0.37 + 0.01(-2) \times 0.6 = 0.358$
$b = 0.16 - 0.02 = 0.14$

Sample 3:

$Z = -0.516(0.4) + 0.08(0.8) + 0.358(0.2) + 0.14$
$= 0.0692$

$\hat{y} = sign(0.0692) = 1$

(correct)

Sample 4:

$Z = 0.0446$ $\hat{y} = sign(Z) = 1$

(missclassified)